Statistical Modeling

David Puelz
The University of Austin

Course goals

- Use supervised learning to build models for prediction and for understanding (causal inference)
- Understand the benefits and limitations of the models we build
- Given a new situation, select an appropriate model, carry it out, and effectively communicate the results
- This course focuses on practice and implementation. But ... we will pick up theoretical details when necessary.

Who am I?

- **Educator:** Teaching at UATX since 2024, UT Austin from 2020-2024, graduate student from 2013-2018. Previously at the University of Chicago Booth School of Business and Goldman Sachs.
- **Researcher:** I develop new statistical/machine learning methods for analyzing complex data, with applications in policy, economics, and the social sciences.
- **Human:** Grew up in Dallas. Husband and dad with 2 girls + 1 boy + 1 dog. I like working out, smoking meat, golf, and cycling.

Introduction to the course

Course logistics

Supervised learning

Building intuition: k-nearest neighbors

The bias-variance tradeoff

Answer two questions for me:

What is one thing you're *excited* about for this class?

What is one thing you're *nervous or concerned* about for this class?

Pass around your answers. Another student will read your responses, so everyone stays anonymous.

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- Access at `uaustin.populiweb.com`
- Access at `https://github.com/dpuelz/StatisticalModeling`
- Github pages will contain schedule, homeworks, data, code, ...
- **Github will be our main landing page**. Populi will be used for grades and administrative tasks.

Class structure

- Understanding the concepts really only comes from practice

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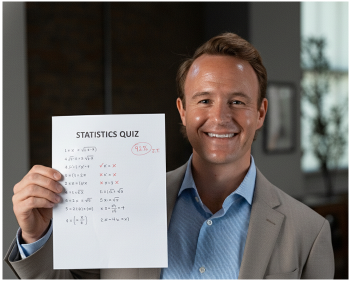
- Class time will be mostly lecture and discussion. We will also have "in-class reps" on Quiz reviews and exercises.

Quizzes

- 5 quizzes, taken during class on the 2nd, 4th, 6th, 8th, and 10th Friday



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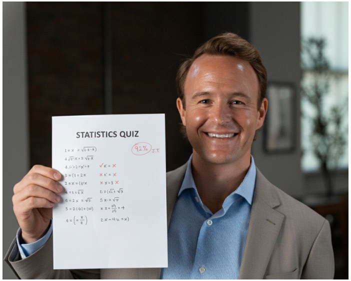

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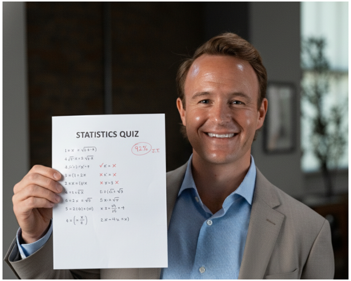
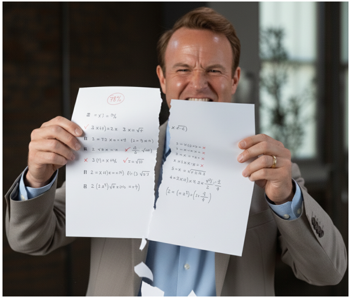
- Assessing your knowledge of lecture material up to that point
- Concepts assessed will come directly from the homeworks



Homework

- Why homework?



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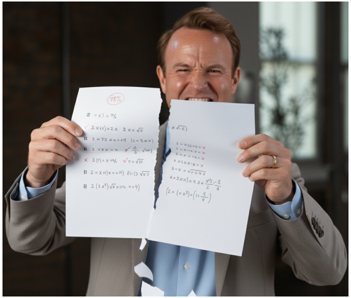

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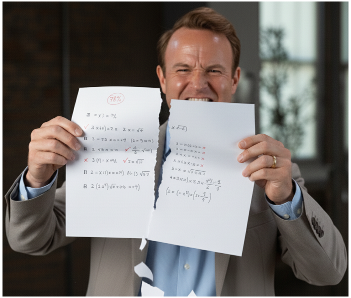

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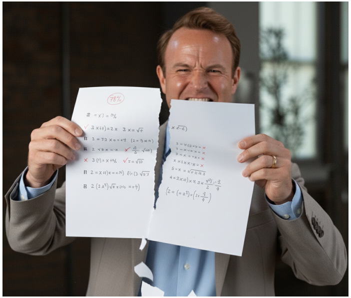
- **Homeworks can be completed individually or in groups of up to three people.** They must be turned in by the start of each Friday class. Names of all group members must be included on the writeups.
- I expect professional writeups in a markdown format (Rmd) or (Jupyter)



A pedagogical note on AI

LLMs can be useful for the homeworks and learning concepts. I encourage their use! And ... I will not police their use. I only ask that you cite when LLMs are used.

However, if overused & relied on too heavily, you will not learn anything.

The quizzes are designed from the homeworks and will assess how well you understand the concepts. **Through the quizzes, you will get feedback quickly on whether you've overused AI!** Use this feedback to optimize your learning with AI.

Exam

- The exam will be during the 11th week (final exam week). There are no make-ups or alternative dates, so please plan accordingly!
- It will be a free response, written exam taken with pen and paper (exactly like our quizzes)

Grading

Homework	**20%**
Exam	**30%**
Quizzes	**50%**

Statistical computing

- We will use R for statistical analysis throughout the course
- You are definitely welcome to use Python, but the course materials will (mostly) be in R
- This is industrial-strength, state-of-the-art, and free software for statistical computing
- We will access R through the RStudio graphical interface; make sure both are installed on your laptop and bring it to every class

Getting help

- My office hours
- Schedule an one-on-one time with me to chat (Zoom or in-person)

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What is supervised learning?

- Simply put, the goal is to predict a target variable Y with input variables X
- In data mining terminology, this is known as supervised learning
- We observe training data: $\{(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)\}$

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Tangent–what do you think **un**supervised learning is?

In general, a useful way to think about it is that Y and X are related in the following way:

$$Y_i = f(X_i) + \epsilon_i$$

The supervised learning framework

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- $f(X_i)$ is the systematic part (the signal)
- ϵ_i is the random error (the noise)
- The main purpose is to *learn or estimate* $f(\cdot)$ from data

Two types of problems

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Today we'll focus on regression, but the same principles apply to classification

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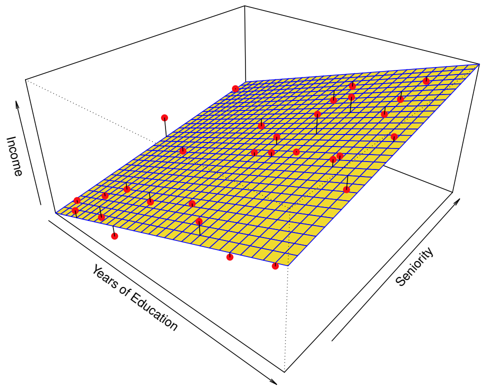
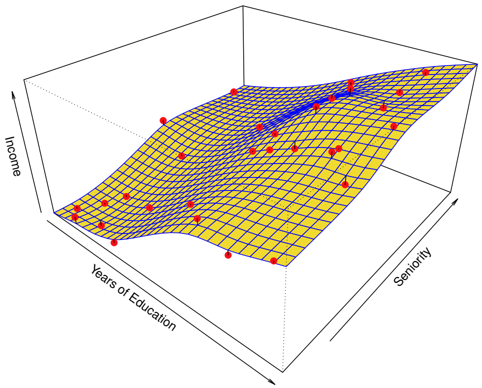
   - Nonparametric model: Let the data determine the form of f (e.g., k-nearest neighbors)
3. Evaluate performance on testing data and *adjust*

Parametric vs. nonparametric



Parametric



Nonparametric

- **Parametric**: Restrictive assumptions, simple interpretation
- **Nonparametric**: Flexible assumptions, complex interpretation

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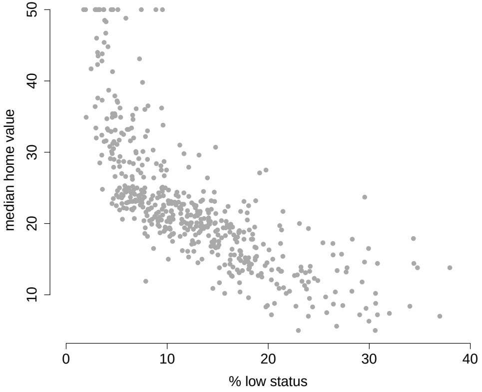
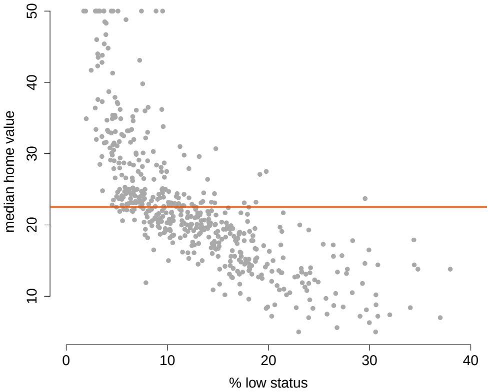
Boston housing example

- Predict median home value with percent low economic status
- What should $f(\cdot)$ be?



What should $f(\cdot)$ be?

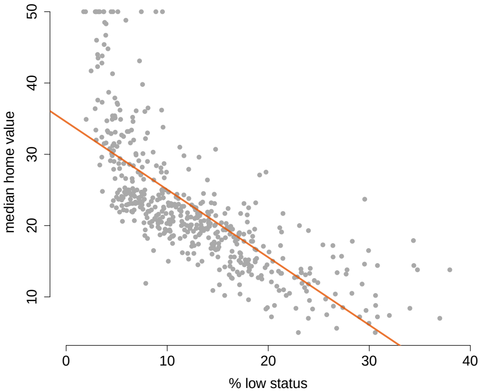
How about this? (Parametric: $Y = \mu + \epsilon$)



If % low status = 30, what is the prediction?

Or maybe this?

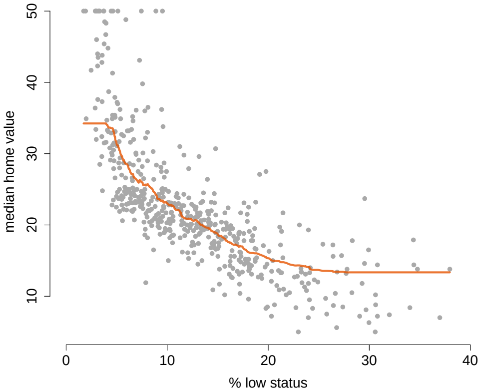
Linear model!



If % low status = 30, what is the prediction?

Or even this?

(k-NN with $k = 100$)



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k-nearest neighbors: How it works

Prediction at point x: $\hat{f}(x) =$ average of k nearest points around x

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- Points that are closer to where we want to predict should be more relevant
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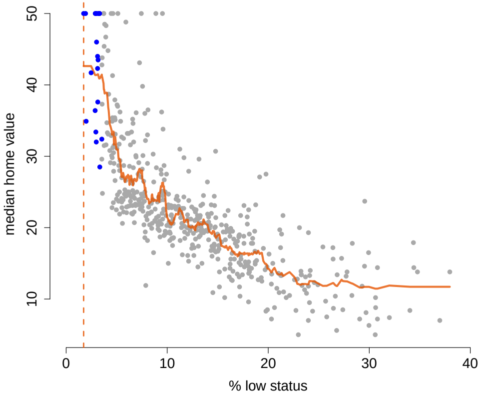
Let's see how this works with $k = 20$...

k-NN with $k = 20$



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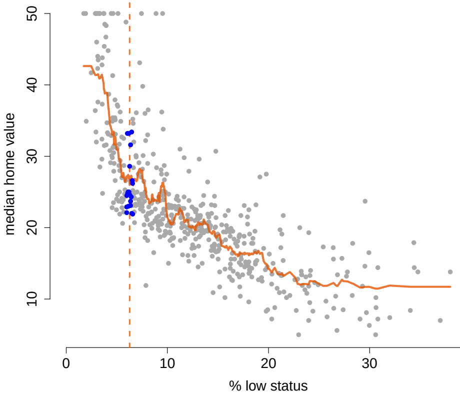

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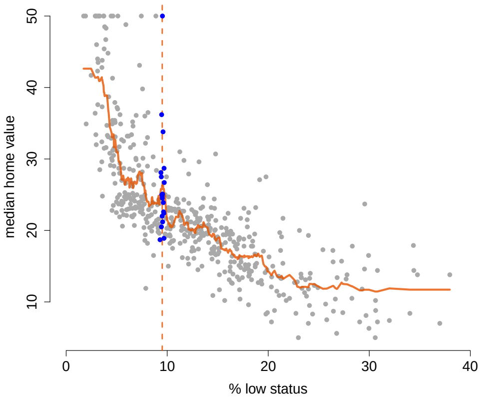

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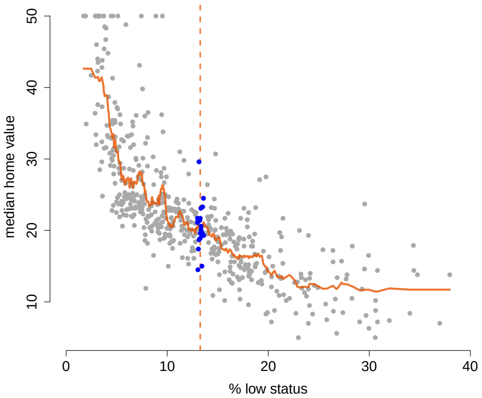

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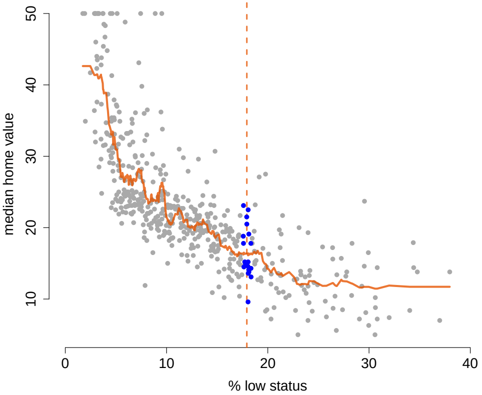

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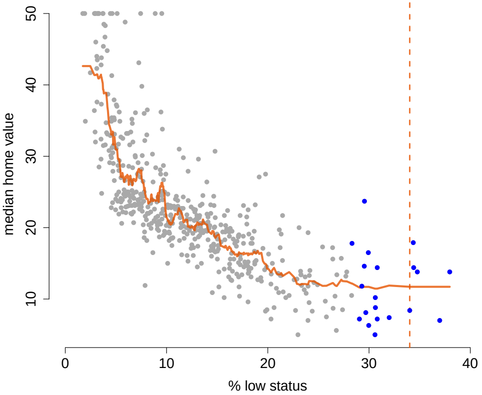
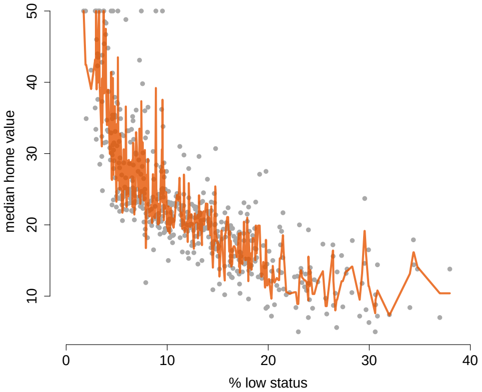
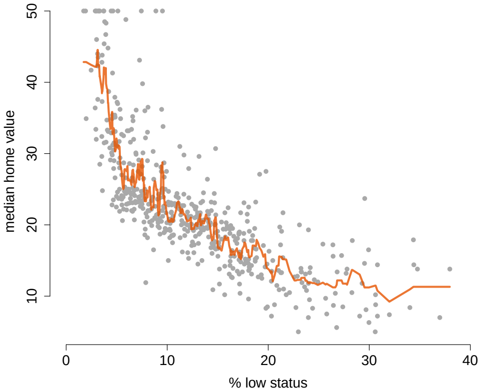


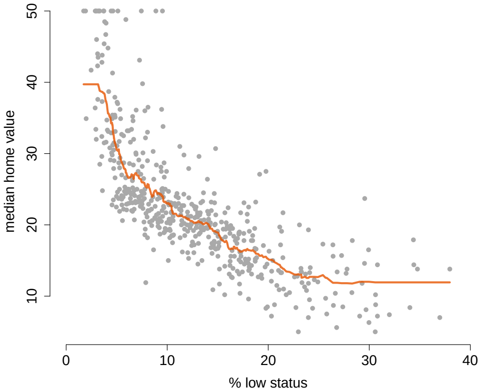
Why not choose $k = 2$?



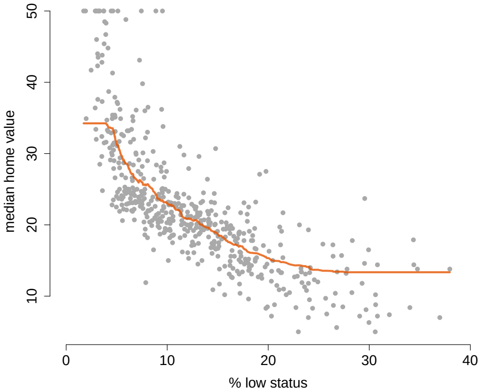
Or $k = 10$?



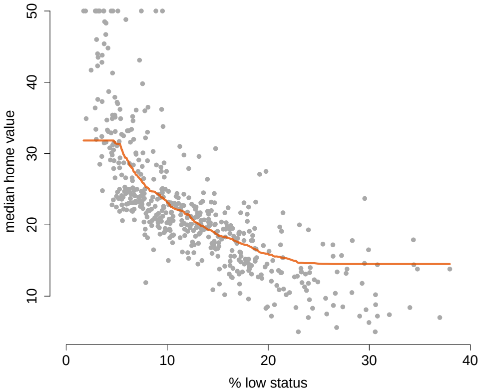
Or $k = 50$?



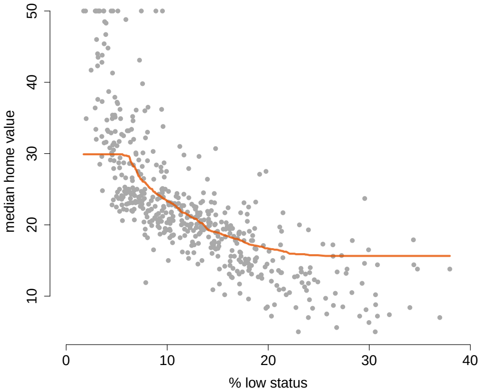
Or $k = 100$?



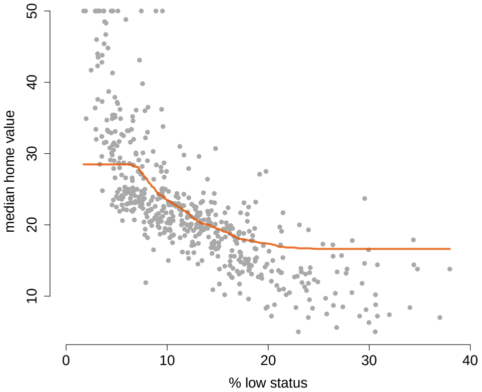
Or $k = 150$?



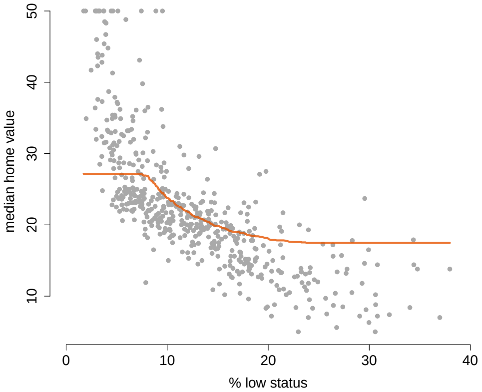
Or $k = 200$?



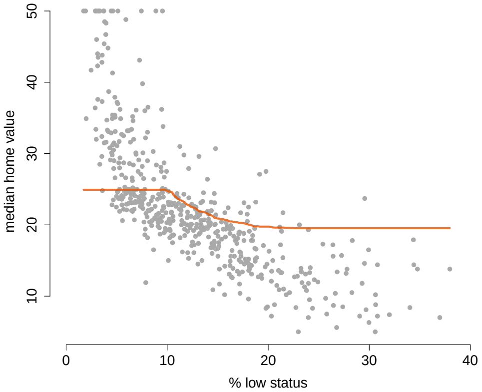
Or $k = 250$?



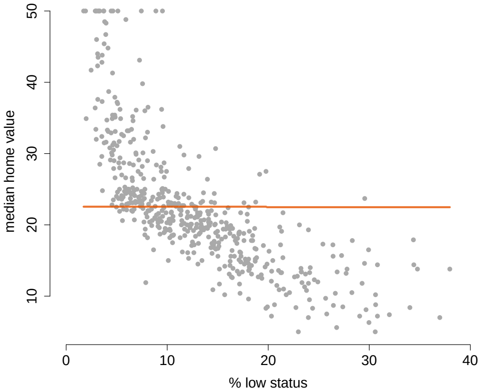
Or $k = 300$?



Or $k = 400$?



Or $k = 505$ (all data)?



Measuring prediction accuracy

- **In-sample error**: How well does the model fit the training data?

$$\mathsf{RMSE}_{\mathsf{in}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}[Y_i - \hat{f}(X_i)]^2}$$

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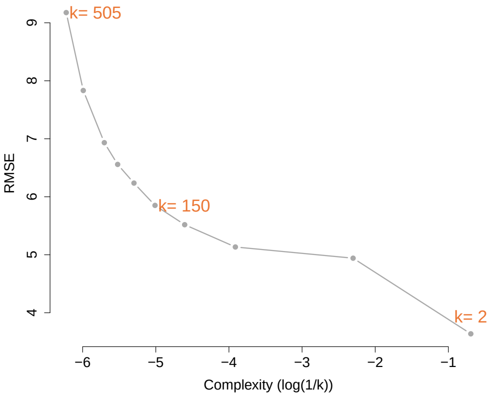
- We care about out-of-sample performance!

In-sample RMSE



It looks like $k = 2$ is the best. Should we choose this model?

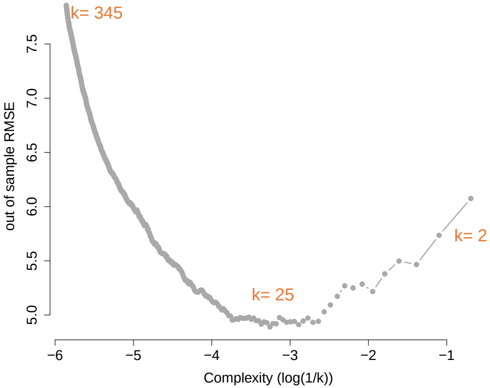
We care about out-of-sample performance!

- Suppose we have m additional observations (X_i^o, Y_i^o) that we did not use to fit the model
- This is called the validation set (or hold-out set, or test set)
- We evaluate with out-of-sample RMSE:

$$\text{RMSE}^o = \sqrt{\frac{1}{m}\sum_{i=1}^{m}[Y_i^o - \hat{f}(X_i^o)]^2}$$

Out-of-sample RMSE

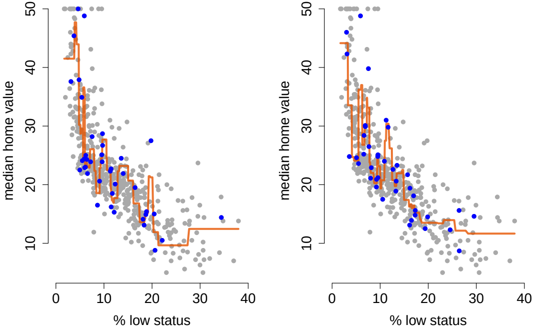
Fit each model on training set (size 400). Test on validation set (size 106).



Now $k \approx 46$ looks like the best choice! Not too simple, but not too complex!

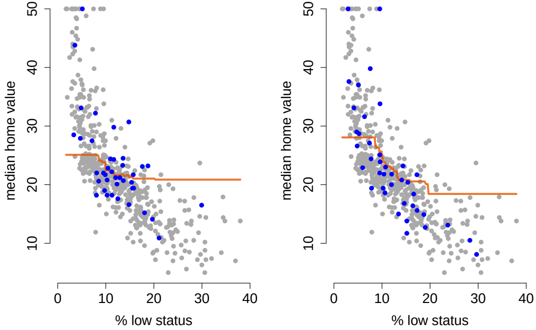
Visualizing bias and variance

$k = 2$: Low bias, high variance



Training set of size 40

$k = 25$: High bias, low variance



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A fundamental challenge

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Our goal: Find the sweet spot in the bias-variance tradeoff!

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  - May not capture all patterns $\rightarrow$ high bias
  - But predictions are stable $\rightarrow$ low variance
- **Complex models** (e.g., small k in k-NN):
  - Can capture complex patterns $\rightarrow$ low bias
  - But predictions vary a lot $\rightarrow$ high variance

The mathematical decomposition

For a prediction at point x, the mean squared error can be decomposed:

$$\mathsf{MSE} = \mathsf{E}[(f(x) - \hat{f}(x))^2] = \underbrace{[\mathsf{E}[\hat{f}(x)] - f(x)]^2}_{\text{Bias}^2} + \underbrace{\mathsf{Var}[\hat{f}(x)]}_{\text{Variance}}$$

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- Bias^2: How far off our average prediction is from the truth

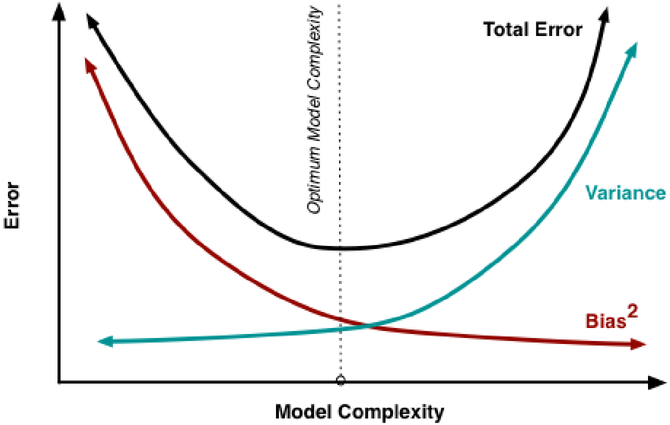
The mathematical decomposition

For a prediction at point x, the mean squared error can be decomposed:

$$\mathsf{MSE} = \mathsf{E}[(f(x) - \hat{f}(x))^2] = \underbrace{[\mathsf{E}[\hat{f}(x)] - f(x)]^2}_{\text{Bias}^2} + \underbrace{\mathsf{Var}[\hat{f}(x)]}_{\text{Variance}}$$

- Bias^2: How far off our average prediction is from the truth
- Variance: How much our predictions vary

Visualizing the tradeoff



- As model complexity increases, bias decreases but variance increases
- The optimal model complexity minimizes the total error

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- The bias-variance tradeoff will continue to guide our thinking throughout the course!

Extra Slides

Deriving the decomposition

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Squaring both sides gives squared terms plus cross terms:

$$(Y - \hat{f}(X))^2 = [f(X) - \mathsf{E}[\hat{f}(X)]]^2 + [\mathsf{E}[\hat{f}(X)] - \hat{f}(X)]^2 + \epsilon^2 + (\text{cross terms})$$

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For the MSE $\mathsf{E}[(f(X) - \hat{f}(X))^2]$, the cross terms vanish, leaving only the two squared terms: bias^2 and variance.